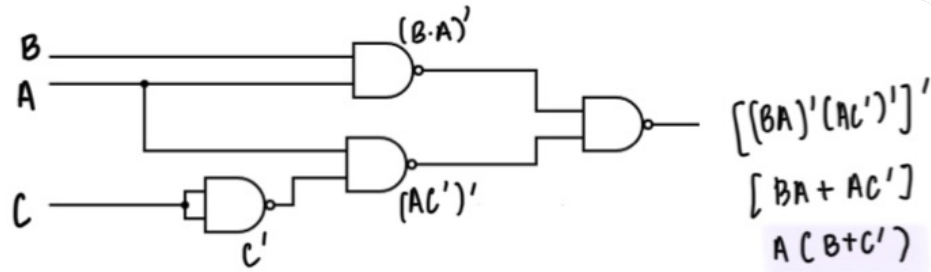


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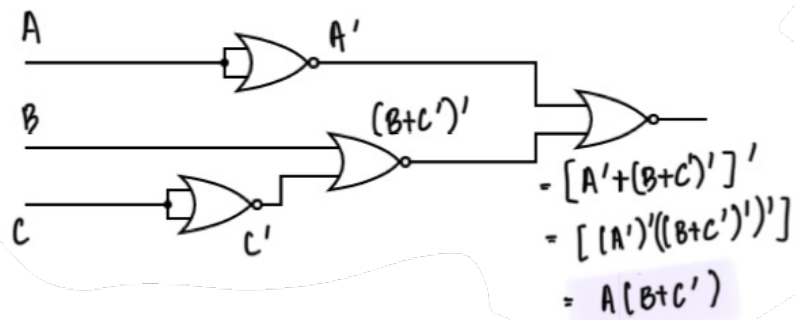
Assignment 2-Part B Solution Manual
 Spring 2026

Q8)

a)



b)



Q9)

a)

CD \ AB	00	01	11	10
00	1 0	1 1	0 3	1 2
01	1 4	1 5	0 7	1 6
11	1 12	1 13	0 15	1 14
10	1 8	1 9	0 11	0 10

Simplified Boolean Expression

$$A'D' + BD' + C'$$

b)

CD AB	00	01	11	10
00	1 0	0 1	0 3	1 2
01	1 4	0 5	0 7	1 6
11	1 12	0 13	0 15	1 14
10	1 8	0 9	0 11	1 10

Simplified Boolean Expression

(D')

c)

CD AB	00	01	11	10
00	1 0	0 1	0 3	1 2
01	0 4	0 5	0 7	0 6
11	0 12	0 13	0 15	0 14
10	1 8	0 9	0 11	1 10

Simplified Boolean Expression

B'D'

Q10)

CD AB	00	01	11	10
00	X 0	1 1	1 3	X 2
01	0 4	X 5	1 7	0 6
11	0 12	0 13	1 15	0 14
10	0 8	0 9	1 11	0 10

Simplified Boolean Expression

A'D + CD

Q11)

Simplification Using Kmap

a)

CD \ AB	00	01	11	10
00	1 0	1 1	1 3	1 2
01	0 4	0 5	0 7	0 6
11	0 12	0 13	0 15	0 14
10	1 8	1 9	1 11	1 10

Simplified Boolean Expression

B'

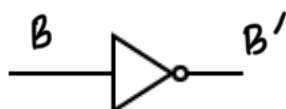
Simplification using boolean algebra

$$\begin{aligned}
 F &= (A' \cdot B' \cdot C') + (A' \cdot B' \cdot C) + (A \cdot B' \cdot C') + (A \cdot B' \cdot C) \\
 &= (A' \cdot B' \cdot (C' + C)) + (A \cdot B' \cdot C') + (A \cdot B' \cdot C) && \text{[Distributive Law]} \\
 &= (A' \cdot B' \cdot 1) + (A \cdot B' \cdot C') + (A \cdot B' \cdot C) && \text{[Complement Law: } C + C' = 1\text{]} \\
 &= A' \cdot B' + (A \cdot B' \cdot C') + (A \cdot B' \cdot C) && \text{[Identity Law]} \\
 &= A' \cdot B' + A \cdot B' \cdot (C' + C) && \text{[Distributive Law]} \\
 &= A' \cdot B' + A \cdot B' \cdot 1 && \text{[Complement Law: } C + C' = 1\text{]} \\
 &= A' \cdot B' + A \cdot B' && \text{[Identity Law]} \\
 &= B' \cdot (A' + A) && \text{[Distributive Law]} \\
 &= B' \cdot 1 && \text{[Complement Law: } A + A' = 1\text{]} \\
 &= B' && \text{[Identity Law]}
 \end{aligned}$$

b)

The entire large expression represents a NOT gate (inverter) on input B. The other inputs (A and C) don't matter

c)



Q12)

a)

X	Y	Z	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

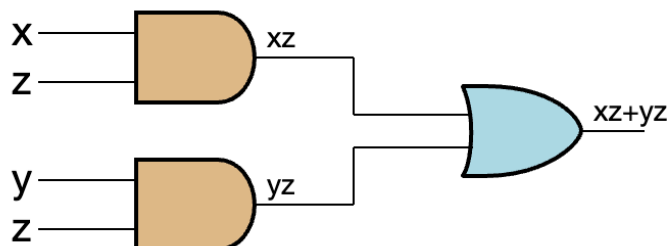
b)

$\begin{matrix} x/y \\ \swarrow \end{matrix}$ Z	0	1
00	0 0	0 1
01	0 2	1 3
11	0 6	1 7
10	0 4	1 5

Simplified Boolean Expression

$$XZ + YZ$$

c)



Q13)

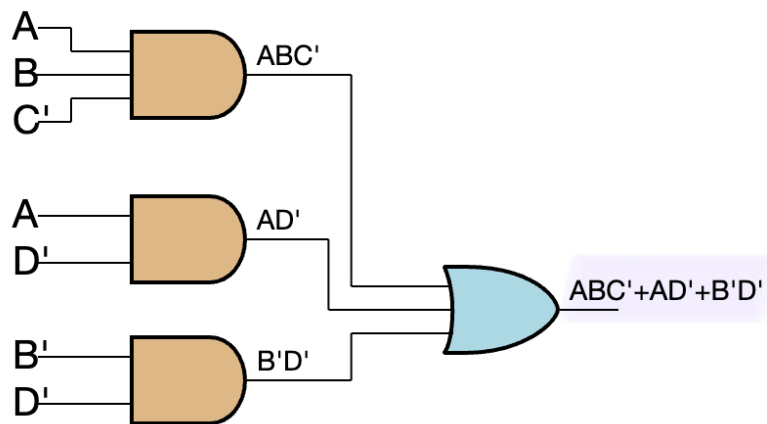
a)

CD \ AB	00	01	11	10
00	1 0	X 1	0 3	1 2
01	0 4	X 5	X 7	0 6
11	1 12	1 13	0 15	1 14
10	1 8	0 9	0 11	1 10

Simplified Boolean Expression

$$ABC' + AD' + B'D'$$

b)



c)

